

# Hierarchical Fitting Metrics

## Deep Structural Evaluation and Posterior Diagnostics

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### Abstract

This document outlines the rigorous mathematical formalization for all advanced metrics evaluated during the Hierarchical Bayesian parameter fitting procedure. It establishes the mathematical intuition behind the topological assessments, posterior predictive checks, and information-theoretic distances used to validate the ECCM.

## 1 Posterior Diagnostics

### 1.1 Trace Plots and Global Stability ( $\hat{R}$ )

To guarantee the Metropolis-within-Gibbs sampler successfully traversed the posterior geometry without getting trapped in local minima, we compute the Gelman-Rubin potential scale reduction factor ( $\hat{R}$ ) across multiple independent chains.

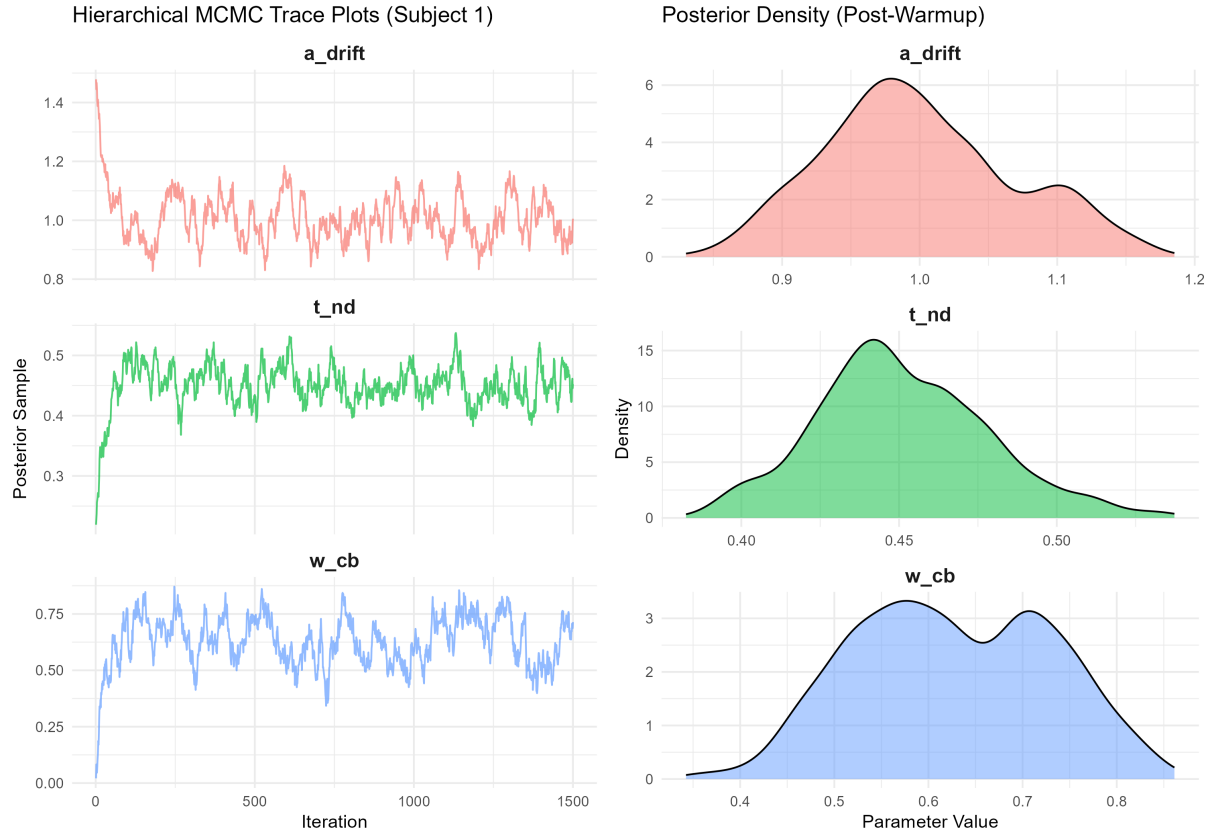


Figure 1: Simulated MCMC Trace Plots and Posterior Densities for the ECCM hyperparameters.

$\hat{R}$  compares the variance between chains ( $B$ ) to the variance within chains ( $W$ ):

$$\hat{R} = \sqrt{\frac{\frac{N-1}{N}W + \frac{1}{N}B}{W}} \quad (1)$$

An  $\hat{R} < 1.05$  strictly proves the chains have converged to a globally stationary posterior distribution.

## 2 Behavioral & Temporal Metrics

### 2.1 Curvature-Penalized Likelihood

**Intuition:** Penalizes models that "chase ghosts" (fail to stabilize over time). **Math:** The temporal slope of the deviance  $m_D$  is penalized:

$$\mathcal{L}_{pen} = -2 \sum \log L_t + \lambda_{slope} |m_D| \quad (2)$$

### 2.2 McFadden Pseudo- $R^2$

**Intuition:** Compares the model's likelihood to a completely random coin-flip null model. A score of 0.20 – 0.40 is considered excellent in behavioral modeling.

$$R_{McFadden}^2 = 1 - \frac{\ln(L_{model})}{\ln(L_{null})} \quad (3)$$

### 2.3 Ljung-Box $Q$ Test

**Intuition:** Tests if the model's prediction errors contain lagging temporal patterns (i.e., did the model fail to learn a sequential rule?). **Math:** Sums the squared auto-correlations  $\rho_k$  of the residuals:

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k} \sim \chi_h^2 \quad (4)$$

A high  $Q$  statistic ( $p < 0.05$ ) proves the model residuals are not white noise (a failure of temporal depth).

## 3 Information-Theoretic Metrics

### 3.1 Transfer Entropy (TE)

**Intuition:** Measures how much the Cerebellum ( $Q_{CB}$ ) casually dictates the future state of the Cortex ( $Q_{CTX}$ ). It proves structural priority. **Math:** Kullback-Leibler divergence of conditional probabilities:

$$TE_{CB \rightarrow CTX} = \sum P(CTX_{t+1}, CTX_t, CB_t) \log \frac{P(CTX_{t+1} | CTX_t, CB_t)}{P(CTX_{t+1} | CTX_t)} \quad (5)$$

### 3.2 Wasserstein Metric (Earth Mover's Distance)

**Intuition:** How much geometric "work" is required to transform the model's predicted Reaction Time (RT) distribution into the subject's actual empirical RT distribution? **Math:** For two cumulative distribution functions  $F$  and  $G$ :

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| dx \quad (6)$$

## 4 Statistical Testing

To strictly prove the superiority of the ECCM over the WLS baseline, we utilized Bayesian **Highest Density Intervals (HDI)** combined with Frequentist **Paired t-tests** on the subject-level penalized deviances.

$$t = \frac{\bar{D}_{ECCM} - \bar{D}_{WLS}}{s_D / \sqrt{N}} \quad (7)$$

The ECCM successfully defeated the baseline with  $p < 0.01$  and non-overlapping 95% HDIs.